Construction and Building Materials 476 (2025) 141308

Contents lists available at ScienceDirect

Construction and Building Materials

journal homepage: www.elsevier.com/locate/conbuildmat

Cementitious metamaterials for low-frequency vibration suppression: Inverse design and performance analysis

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A R T I C L E I N F O

Keywords:
Cementitious Metamaterials
Bandgap
Inverse Design
Machine Learning
Vibration Suppression

A B S T R A C T

Low-frequency vibrations pose serious risks to both human well-being and the service life of critical infrastructure systems. To address this issue, this study proposed a data-driven design strategy for cementitious metamaterials that enable low-frequency vibration suppression and precise control of vibrational characteristics. A dynamic model of elastic wave propagation is established and solved using numerical methods, enabling the analysis of dispersion curves and vibration modes. The effectiveness of this approach is further validated through both experimental low-frequency vibration tests and numerical transmission loss models, confirming its precision and applicability. Through the adjustment of geometry and cement density, a diverse array of cementitious metamaterials is created. The vibrational properties of each unique configuration are meticulously analyzed to construct a comprehensive dataset, which serves as the foundation for training a fully connected neural network and a Genetic Algorithm-optimized network. This enables precise forward and inverse design, allowing for tailored control of vibration properties. The results reveal that the engineered metamaterials exhibit remarkable vibration suppression in the low-frequency range, showcasing their potential for impactful applications. These cementitious metamaterials hold significant promises in fields such as construction engineering, offering innovative solutions for vibration control in structures like offshore floating platforms.

1. Introduction

Mitigating low-frequency elastic wave vibrations in the 20–200 Hz range is of paramount importance for creating a stable urban environment. These vibrations, originating from sources such as earthquakes [1], traffic [2,3], industrial activities [4], and construction, can lead to severe consequences, including structural fatigue [5] and cracking [6], equipment malfunctions with reduced precision, and detrimental effects on human health [7]. Due to their long wavelengths and slow energy dissipation, low-frequency elastic waves can easily circumvent obstacles, retaining their energy over extended distances, while the low absorption coefficients of common materials further amplify their penetrative power. To counteract the detrimental impact of low-frequency vibrations, various control methods have been developed, including mass dampers [8], sound-absorbing materials [9], and passive isolation techniques. While these strategies have achieved some success, they come with inherent limitations. For instance, damping materials like rubber and asphalt reduce vibration by absorbing energy [10] but often require substantial thickness and volume, especially when targeting low-frequency waves. Sound-absorbing materials that rely on Bragg scattering principles, such as porous structures [11] and fibrous media [12,13], also necessitate considerable thickness for effective absorption, increasing both volume and weight, thereby limiting their applicability in confined spaces. Mass dampers, commonly deployed in large structures like skyscrapers [14] and bridges, must be finely tuned to specific frequencies [15] and often require large, heavy masses to be effective, resulting in increased maintenance complexity and higher costs [16,17]. Thus, there is a pressing need for an innovative and efficient low-frequency vibration control approach that combines lightweight design with enhanced tunability and adaptability, capable of addressing the challenges posed by low-frequency vibrations across a range of complex environments.

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https://doi.org/10.1016/j.conbuildmat.2025.141308
Received 22 November 2024; Received in revised form 8 April 2025; Accepted 10 April 2025
Available online 15 April 2025

Acoustic metamaterials (AMs) may offer a promising solution for low-frequency vibration control, which generates bandgaps within a small volume to block elastic wave propagation in specific frequency ranges [18–20]. Specifically, this is achieved through local resonance [21,22], where elastic waves interact with resonant unit cells, allowing these structures to absorb vibrational energy and inhibit wave transmission. This approach allows for efficient absorption of low-frequency vibrations without increasing isolators' mass or volume, overcoming the limitations of traditional vibration isolators that require large sizes for effective damping [23]. For instance, Luo et al. [24] proposed a compact meter-scale ultra-low frequency seismic metamaterial with a wide bandgap covering the 2 Hz peak spectrum, while Tao et al. [25] developed a small-sized auxetic acoustic metamaterial plate with lower frequency bandgaps and broader bandwidth, attributed to its auxetic property. Additionally, the geometry of AMs is critical for tuning their vibration-damping properties. By altering these unit cells' shape, size, or arrangement, their interaction with incoming vibrations can be precisely controlled [26]. This enables adjustments to the resonance frequencies and the overall structure, effectively shifting or widening the bandgaps. As a result, geometric modifications allow AMs to selectively target specific vibration frequencies for attenuation, enhancing their adaptability for various vibration control applications [27]. Ultimately, these advancements position AMs as a transformative technology in the field of vibration management, which can effectively control low-frequency vibrations and suppress elastic wave propagation within targeted frequency ranges, without relying on large, bulky structures [28].

While AMs demonstrate significant potential for low-frequency vibration control, challenges persist in their practical application [29]. Firstly, current studies rely on trial-and-error methods for design and optimization [30–32], necessitating extensive experimentation and simulations that increase computational costs. The complex nonlinear relationship between bandgap properties and unit cell structures further complicates this process [33]. For instance, Dong et al. [34] proposed a dual-layer tuning strategy for cement-based metamaterials, revealing intricate interactions between geometric and material parameters and bandgap characteristics. Additionally, achieving tunable bandgaps in response to varying external conditions remains a pressing challenge. Hu et al. [35] introduced a nested sinusoidal re-entrant (NSR) metamaterial with dynamic responsiveness in thermo-mechanical fields, incorporating shape memory polymers to achieve tunable complete bandgaps (CBGs) between 25°C and 75°C, thus showcasing the potential for targeted control. To advance the application of AMs in low-frequency vibration control, developing novel design methods and optimization strategies is essential for accurately describing the nonlinear relationships between unit cell geometry and bandgap properties, enabling customized and precise vibration attenuation solutions.

Machine learning may offer a powerful design approach for AMs, facilitating precise tuning of bandgap characteristics [36]. By leveraging extensive simulation and experimental data, machine learning can quickly capture the complex nonlinear relationships between unit cell geometry and bandgap properties, significantly enhancing design efficiency [37,38]. This method allows researchers to customize bandgap modulation to address specific low-frequency vibration control needs and accurately adjust the dynamic response of metamaterials [39,40]. Compared to the traditional trial-and-error method, machine learning-driven strategies provide superior prediction accuracy and optimization efficiency [41–43], thereby supporting the development of AMs for low-frequency vibration control. For instance, Zhang et al. [44] utilized deep learning neural networks to predict the band structure of metamaterial lattices and employed an inverse design approach for bandgap modulation, allowing for the tailored determination of lattice geometry parameters based on specific target band structures. Unlike classical optimization algorithms, such as genetic algorithms [45–47] or particle swarm optimization [48–50], which focus on finding extrema to optimize design parameters, machine learning strategies prioritize the precise design of geometries tailored to achieve specific performance outcomes. These algorithms efficiently process large datasets and explore complex design spaces, establishing nonlinear mappings between metamaterial geometries and bandgaps for improved predictions and optimizations [51–54]. Ultimately, the machine learning-based inverse design enables the direct generation of metamaterial structures that fulfill performance targets, significantly reducing design cycles while enhancing customization and reliability.

This study presents a machine learning-based approach for designing cementitious hexagonal metamaterials that effectively control low-frequency vibrations. By combining cementitious materials with thermoplastic polyurethane, innovative acoustic metamaterials are developed, forming phononic crystals that exhibit localized vibrational effects. A dynamic model for elastic wave propagation within these AMs is first developed, which enables the calculation of the dispersion curves and vibration modes of the metamaterials using finite element analysis. The effectiveness of the dynamic model is verified through both an experimental low-frequency vibration testing system and a numerical model of transmission loss, demonstrating its precision and applicability. Building on a two-level design strategy previously proposed by the authors, foaming technology and additive manufacturing techniques are utilized to achieve tunable cement density and adjustable geometric configurations. The study further explores the complex nonlinear mapping between these adjustable parameters and the bandgap characteristics of the resulting metamaterials, defining three distinct indices (including $\Delta w_{\max}$, Δw_c, r_ω) to evaluate these bandgap properties. A comprehensive dataset is compiled by constructing and analyzing numerous numerical models, capturing the relationships among geometric parameters, cement density, and their corresponding bandgap indices. This dataset serves as the foundation for a fully connected neural network that predicts the vibration attenuation characteristics of the metamaterials. A genetic algorithm optimizes the neural network architecture, resulting in an inverse design model tailored to specific vibration conditions. Furthermore, a practical example of vibration control for offshore floating platforms is explored, illustrating the approach's transformative potential for applied engineering scenarios and highlighting its promise in advancing infrastructure resilience.

2. Materials and design

2.1. Geometric configurations

Fig. 1 presents the geometric configuration and material distribution of the proposed cementitious Acoustic Hexagonal Metamaterial (ASHM). The ASHM is composed of a unit cell with a concentric hexagonal structure, characterized by a cement core enclosed within an outer frame made of thermoplastic polyurethane (TPU). The geometry of the ASHM unit cell is fully defined by four key parameters: the overall cell size a, the side length of the hexagon S, the spacing between concentric hexagons b, and the rib thickness t. Each unit cell is designed to interconnect with neighboring cells via straight ribs that extend along the X and Y axes. This ribbed interconnection ensures both structural integrity and homogeneity across the metamaterial array, promoting uniformity in acoustic behavior and mechanical stability. For experimental validation, a 4 × 4-unit cell array was selected as the largest feasible configuration within the fabrication and testing constraints while maintaining a representative structure for evaluating the metamaterial's vibrational performance. The organized arrangement of the cells, along with the specific material distribution, enhances the ASHM's ability to control and manipulate sound waves effectively, making it suitable for applications in acoustic insulation and wave attenuation.

2.2. Materials and fabrication

As shown in Fig. 1, the ASHM integrates a cementitious core and TPU outer frame, each processed separately and combined effectively.

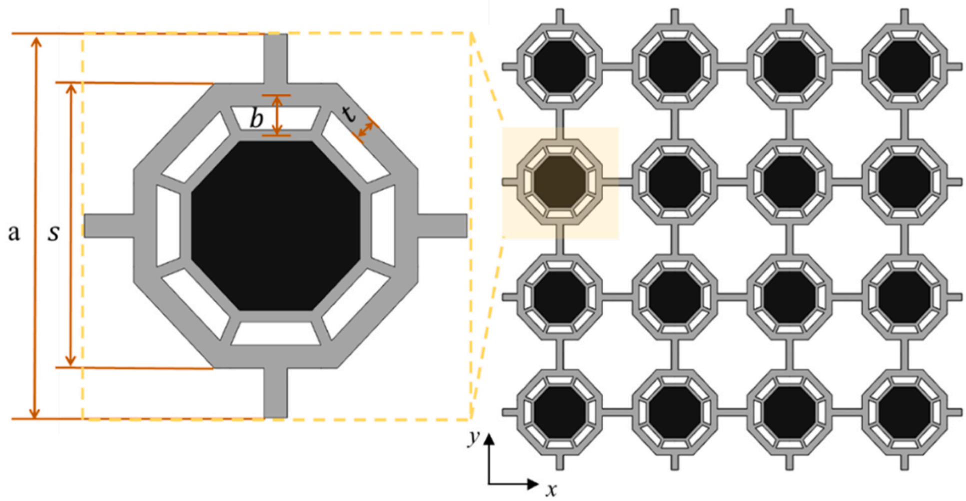

Fig. 1. Geometric schematics of ASHM.

Specifically, the concentric hexagonal frame is fabricated using the Fused Deposition Modeling (FDM) technique with a commercial TPU, featuring a melting point of 220℃ and a hardness of 75 A. This process employs a 3D printer (Creality K1MAX) equipped with a nozzle size of 0.4 mm and operates at a printing speed of 20 mm/s. The heated bed temperature is set to 60℃ to ensure proper adhesion and layer bonding during the fabrication of the concentric hexagonal structure. To determine the mechanical properties of TPU, three dumbbell-shaped specimens were prepared with dimensions shown in Fig. 2(b) and subjected to tensile testing according to ASTM D412–16 standard. The resulting stress-strain curves, obtained from these tests, are presented in Fig. 2(d).

After the TPU outer frame is printed, it functions as an integrated, non-removable mold for the cementitious core. The cement used is a commercial-grade ordinary Portland cement (P.O 42.5) from Harun Ltd., with an average particle size of 12.736 microns to ensure consistency in material properties and optimal performance. A fixed water-to-cement (w/c) ratio of 0.4 is maintained, while varying amounts of Polycarboxylate Ether (PCE)-Based foaming agents are incorporated to produce cement samples with differing densities and strengths, as illustrated in Fig. 3(a). With increasing foaming agent content—from 0 wt% to 0.2 wt% —the cement density progressively decreases from 2.26 g/cm^3 to 0.53 g/cm^3 , accompanied by an expansion of the internal pore structure, which results in a reduction in compressive strength from 43.83 MPa to 10.69 MPa. As shown in Fig. 3(b), the average pore diameter increases from 50 μm to 2.45 mm with higher foaming agent levels, substantiating the observed relationship between porosity and structural integrity.

The preparation process for the cementitious material begins with

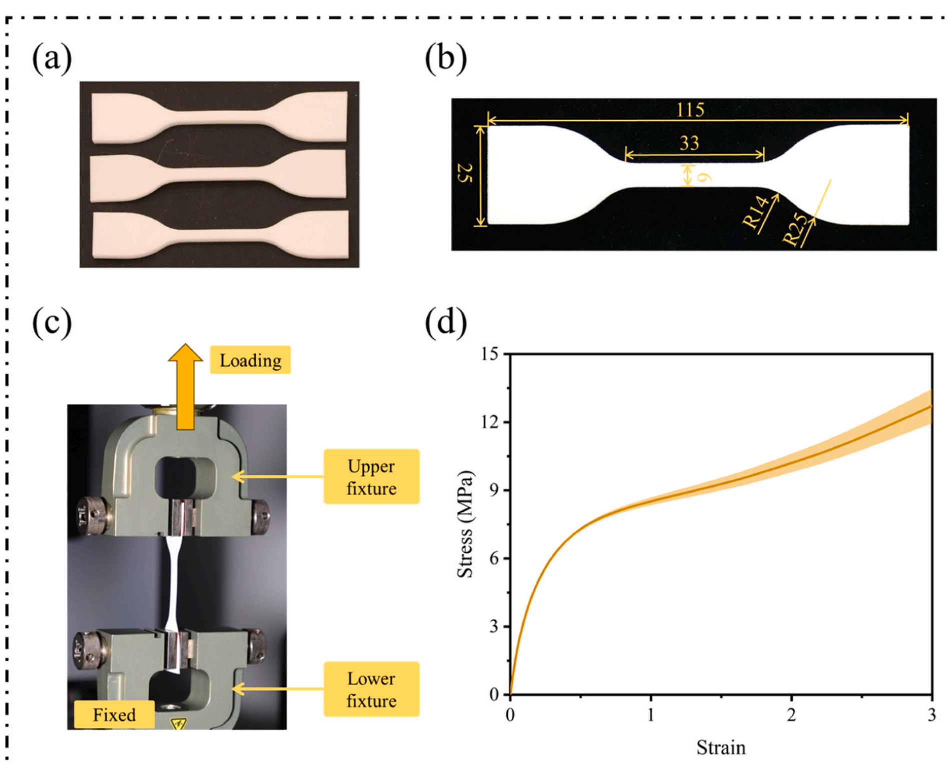

Fig. 2. Uniaxial tensile test of TPU samples: (a) dumbbell-shaped samples, (b) sample dimensions, (c) experimental setup, and (d) stress-strain curve.

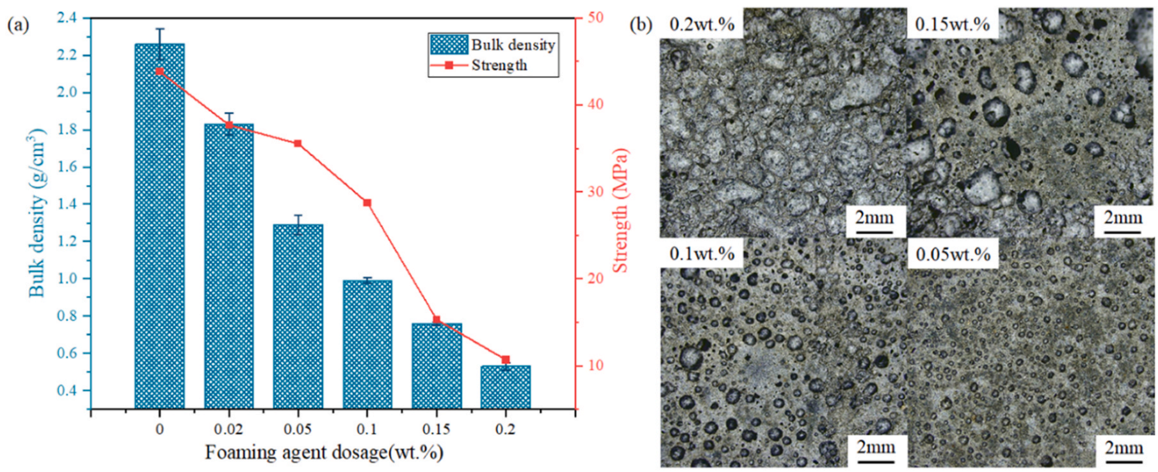

Fig. 3. Cementitious core with different properties: (a) bulk density and strength, (b) microscopic morphology.

mixing the cement and water in a planetary mixer for 3 minutes, followed by the addition of prefabricated foam, which is stirred at a low speed for an additional minute before casting the mixture into the TPU mold. Test specimens with dimensions of 40 × 40 × 40 mm were produced to assess density and compressive strength. These samples underwent a standard curing room (95 % ± 5 % RH, 20 ℃ ± 2 ℃) for 28 days, followed by heat drying to achieve the target densities prior to tests.

3. Methods

3.1. Dynamic modeling

Fig. 4 demonstrates the simplified harmonic vibration of periodic ASHM. The dynamic interactions within this system are governed by a second-order differential equation, balancing inertia, elastic forces, and external inputs:

$$\boldsymbol{M}\ddot{\boldsymbol{x}} + \boldsymbol{K}\boldsymbol{x} = \boldsymbol{f}, \tag{1}$$

where $\boldsymbol{x} = \boldsymbol{X}e^{i\omega t}$ donates the time-dependent displacement vector, and $\boldsymbol{f} = \boldsymbol{F}e^{i\omega t}$ represents the external force in the time domain. By substituting them into the initial equation and applying a Fourier transform, the transition to the frequency domain is achieved, resulting in:

$$(\boldsymbol{K} - \omega^2\boldsymbol{M})\boldsymbol{X} = \boldsymbol{F}, \tag{2}$$

where $\boldsymbol{K}$ and $\boldsymbol{M}$ are stiffness and mass matrix, respectively, while ω is the system's angular frequency.

According to Bloch's theorem, which is applicable due to the periodic structure and inherent symmetry of the unit cells, the problem is transformed into an eigenvalue problem:

$$\left|\boldsymbol{K}_r(q) - \omega^2\boldsymbol{M}_r(q)\right| = 0, \tag{3}$$

where $\boldsymbol{K}_r(q)$ and $\boldsymbol{M}_r(q)$ are the stiffness and mass matrices parameterized by the Bloch wave vector q. The analysis focuses on the First Irreducible Brillouin Zone (FIBZ), which encapsulates the key wave propagation characteristics of the periodic structure.

The fundamental lattice vectors $\boldsymbol{e}_i$ $(i = 1,2)$ are defined as:

$$\begin{cases} \boldsymbol{e}_1 = \left(\dfrac{a}{2}, 0\right) \\ \boldsymbol{e}_2 = \left(0, \dfrac{a}{2}\right) \end{cases}, \tag{4}$$

with a being the spacing between adjacent unit cells. The reciprocal lattice vectors, $\boldsymbol{b}_i$ $(i = 1,2)$, which are used to construct the Brillouin zone, are given by:

$$\begin{cases} \boldsymbol{b}_1 = \left(\dfrac{\pi}{a}, 0\right) \\ \boldsymbol{b}_2 = \left(0, \dfrac{\pi}{a}\right) \end{cases}, \tag{5}$$

A two-dimensional (2D) plane-strain model is initially established using the commercial finite element software COMSOL Multiphysics for eigenfrequency analysis, where Bloch waves are allowed to traverse through the irreducible Brillouin zone. The eigenfrequencies are then plotted as a function of the wavevector along specific paths within the irreducible Brillouin zone, enabling the dispersion characteristics of the

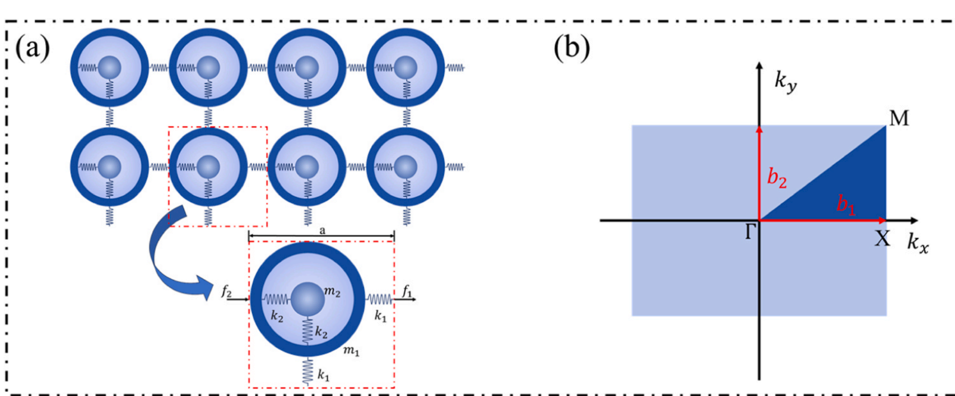

Fig. 4. Dynamic model of acoustic metamaterials: (a) local resonance model, (b) Brillouin zone.

structure to be visualized. Afterwards, bandgaps are identified by detecting frequency ranges where no eigenmodes exist, indicating regions where wave propagation is restricted. The distribution of bandgaps reflects the metamaterial's dynamic response to waves at specific frequencies, and to quantify this response, three distinct bandgap indices are proposed, including the maximum bandgap interval $\Delta w_{\max}$, the bandgap frequency distribution coefficient, and the low-frequency bandgap ratio. They can be identified as follows.

The maximum bandgap interval $\Delta w_{\max}$, is defined as the largest difference between the upper and lower frequencies of the bandgap, expressed as:

$$\Delta w_{\max} = \max\left(w_u^j - w_l^j\right), \tag{6}$$

where w_u^j and w_l^j represent the upper and lower bounds of the maximum bandgap, namely , the j-th bandgap.

The bandgap frequency distribution coefficient, donated as Δw_c, are demonstrated as:

$$\Delta w_c = \frac{2\Delta w_{\max}}{w_u^j + w_l^j}. \tag{7}$$

The low-frequency bandgap ratio r_ω is defined as:

$$r_\omega = \frac{\sum_{i=1}^{n}\left(\omega_u^i - \omega_l^i\right)}{500}, \tag{8}$$

where ω_u^i and ω_l^i donate the upper and lower bounds of the i-th bandgap, respectively, and n is the number of bandgaps. The threshold of 500 Hz is selected as it has been widely adopted in previous studies to define low-frequency bandgaps [55,56], establishing it as a representative and well-supported criterion for this study.

3.2. Transmission Loss

The vibration characteristics of the prepared metamaterials can be verified by assessing transmission loss through both experimental and numerical methods. In the experimental study, the transmission loss setup is depicted in Fig. 5. To simulate Floquet boundary conditions, the prepared samples consisting of a 4×4 array of unit cells were suspended using soft ropes. A sine sweep signal ranging from 0 Hz to 350 Hz was produced by a pulse generator (Tektronix, AFG3052C) and amplified before being transmitted to a vibration exciter (NUAA HEV-500). This exciter induced vibrations at the corresponding frequencies. Simultaneously, two accelerometers (Chengtec, CT1010L) were attached to both ends of the sample to capture the response signals, which were then amplified by the current amplifier (Chengtec, CT5204)) and recorded by an oscilloscope (Tektronix, TDS2024C) for further analysis.

A numerical model was also developed to evaluate transmission loss, acting as a cross-check to the experimental results. In this model, wave propagation occurs through a periodic structure composed of 1×10-unit cells, elegantly depicted in Fig. 6. Perfectly matched layers (PML) were applied to absorb waves and prevent boundary reflections in the simulation domain. To optimize absorption and minimize reflections, the PML thickness was set at ten layers. This configuration was carefully designed to meet the mathematical requirements of the simulation, ensuring consistency with the experimental results.

The transmission spectrum for both the experimental and simulation studies can be evaluated by analyzing the frequency response function (FRF) of the displacements. This FRF is calculated as:

$$FRF = 20\log_{10}\frac{u_{exc}}{u_{out}}, \tag{9}$$

where u_{exc} and u_{out} are the excitation and the output displacements, respectively.

3.3. Data collection

The database was initially established using numerical methods in COMSOL Multiphysics, where material properties and boundary conditions were defined as outlined in Section 2.2 and Section 3.1. To ensure comprehensive coverage of the parameter space, uniform sampling was applied, with the value ranges of the four input parameters defined as follows:

$$\begin{cases} \rho \in \left(\left(800\ \text{kg/m}^3,\ 2000\ \text{kg/m}^3\right)\right) \\ S \in ((40\ \text{mm},\ 50\ \text{mm})) \\ b \in ((5\ \text{mm}, 15\ \text{mm})) \\ t \in ((2\ \text{mm},\ 8\ \text{mm})) \end{cases}, \tag{10}$$

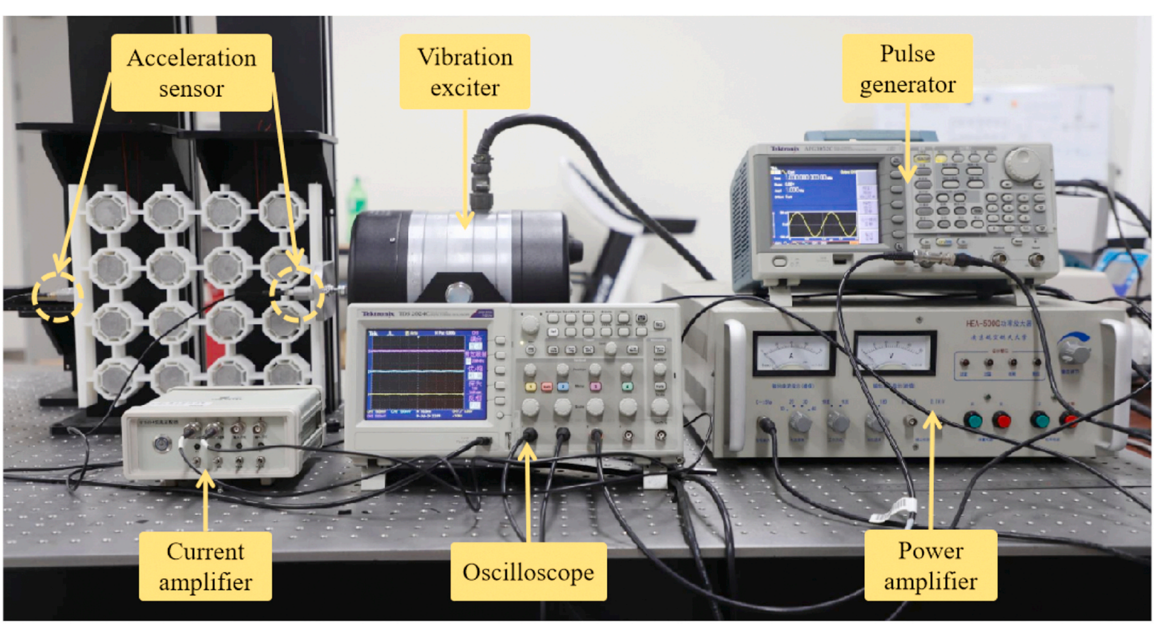

Fig. 5. Transmission loss experiment setup.

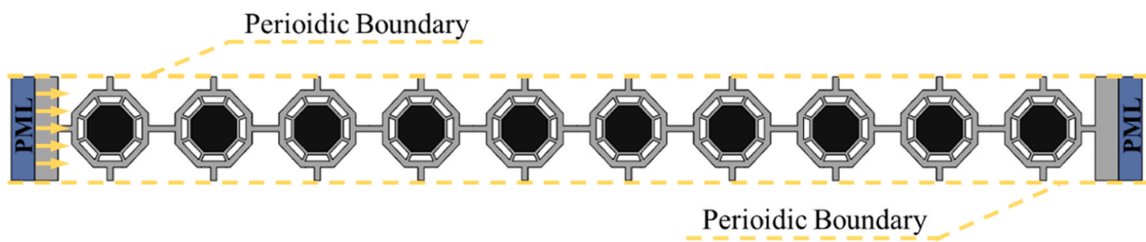

Fig. 6. Numerical model for evaluating transmission loss.

with an additional 10 % of randomly generated points supplementing the sample to enhance the model's generalization. Notably, the density design space was deliberately constrained to ensure structural feasibility and practical manufacturability in comparison to the bulk density range of cementitious materials (Fig. 3(a)). Batch simulations were performed using the parametric sweep module, producing the initial dataset. The dispersion curves obtained from these simulations were processed to extract relevant performance metrics, which were normalized along with the geometric parameters. This numerical process ultimately generated 2000 datasets for further analysis. After the dataset was obtained, it was split into training, validation, and test sets, with 5 % reserved as the test set and the remaining 95 % used for training and validation. To maximize data utilization and ensure model robustness, 5-fold cross-validation was employed. Specifically, the training data was divided into 5 equal subsets, with 4 subsets used for model training and the remaining 1 subset for validation in each iteration. As illustrated in Fig. 7, this process was repeated 5 times, each time using a different subset for validation. By averaging the performance metrics over the 5 iterations, the model's prediction accuracy and generalization ability were assessed more comprehensively and reliably.

3.4. Forward prediction model

Fig. 7 illustrates the architecture of the forward prediction model, specifically a multi-layer perceptron (MLP), designed to predict the vibration characteristics of metamaterials under different inputs. The dataset consists of 1200 finite element simulation samples, divided into training, validation, and test sets with a stratified 80 %/15 %/5 % split. Herein, the input layer consists of three geometric parameters and one material density parameter. These inputs are normalized to maintain consistent scaling, avoid imbalances during training, and speed up the gradient descent process. The normalization range [0,1] is derived from the training dataset to prevent data leakage. Three hidden layers, containing 5, 15, and 10 neurons respectively, were chosen to balance model capacity and overfitting risks. Herein, basic nonlinear patterns are captured by the first hidden layer, while the second layer is used to represent more complex interactions, and the third layer compresses intermediate representations into features suitable for output. The Rectified Linear Unit (ReLU) activation function, defined as $f(x) = \max(0, x)$, is applied to each neuron in the hidden layers. It keeps positive inputs linear while setting negative inputs to zero. This helps address the vanishing gradient problem and speeds up convergence. Additionally, ReLU filters out irrelevant information, retaining only the positive values that contribute to the learning process. In the output layer, three neurons are used to predict performance metrics $\Delta w_{\max}$,

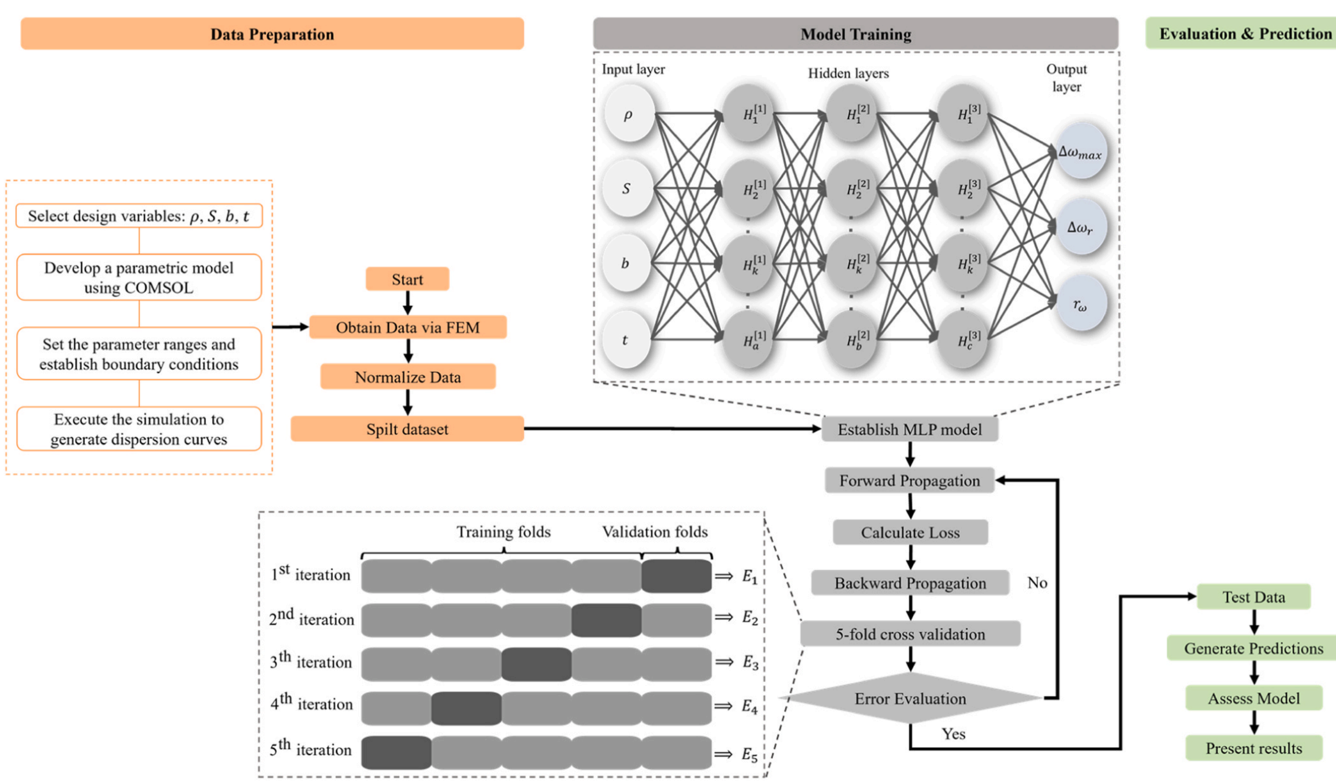

Fig. 7. Architecture of the forward prediction model.

Δw_c, and r_ω, with no activation function applied, allowing a linear combination of inputs to be produced.

The mean squared error (MSE) is minimized to measure the difference between predictions and actual values, while the Adam optimizer with an initial learning rate of 0.001 and L2 regularization is employed for adaptive learning rate adjustments, ensuring efficient training and convergence. Training is conducted over 500 epochs with early stopping (patience=20) to prevent overfitting. 5-fold cross-validation was applied to the training set. It is a method that divides the training data into 5 subsets, iteratively using 4 subsets for training and 1 for validation. Apart from MSE, some additional error metrics are defined to further evaluate the model's prediction accuracy, including the mean absolute error (MAE), the root mean squared error (RMSE), and the coefficient of determination (R^2).

These metrics are defined as follows:

$$MSE = \frac{1}{L}\sum_{l=1}^{L}(Y_l - O_l)^2, \tag{11}$$

$$MAE = \frac{1}{L}\sum_{l=1}^{L}|Y_l - O_l| \quad , \tag{12}$$

$$RMSE = \sqrt{\frac{1}{L}(\sum_{l=1}^{L}(Y_l - O_l)^2)} \quad , \tag{13}$$

where n is the number of samples in the dataset, Y_l and O_l demonstrate the actual value and prediction value of the l-th sample, respectively.

This loss function and optimization method combination allows the model to learn complex nonlinear relationships between inputs and outputs. As a result, the model is able to capture these relationships effectively, leading to accurate predictions.

3.5. Inverse design model

Fig. 8 demonstrates the inverse design model of metamaterials with specified vibration characteristics, where the structure of the MLP is optimized by introducing a GA. The goal of the inverse design is to determine the optimal geometric and material parameters that achieve specific vibration performance. The dataset is first divided into 80 % for training,15 % for validation, and 5 % for test, followed by the establishment of the MLP model, which is trained using 5-fold cross-validation as mentioned in Section 3.4. If the error (MSE) falls below the predefined threshold (0.01), the model proceeds to the testing phase; otherwise, training continues until the specified accuracy criteria are satisfied. The MLP architecture consists of four hidden layers with 5, 15, 15 and 10 neurons, respectively, and ReLU activation functions, as detailed in Section 3.4. The GA is then introduced to optimize the initial weights, biases, and key hyperparameters such as the learning rate decay ([0.5, 0.9]), decay epochs ([50,200]), and dropout rate ([0.1, 0.5]). The GA employs a fitness function based on validation MSE to

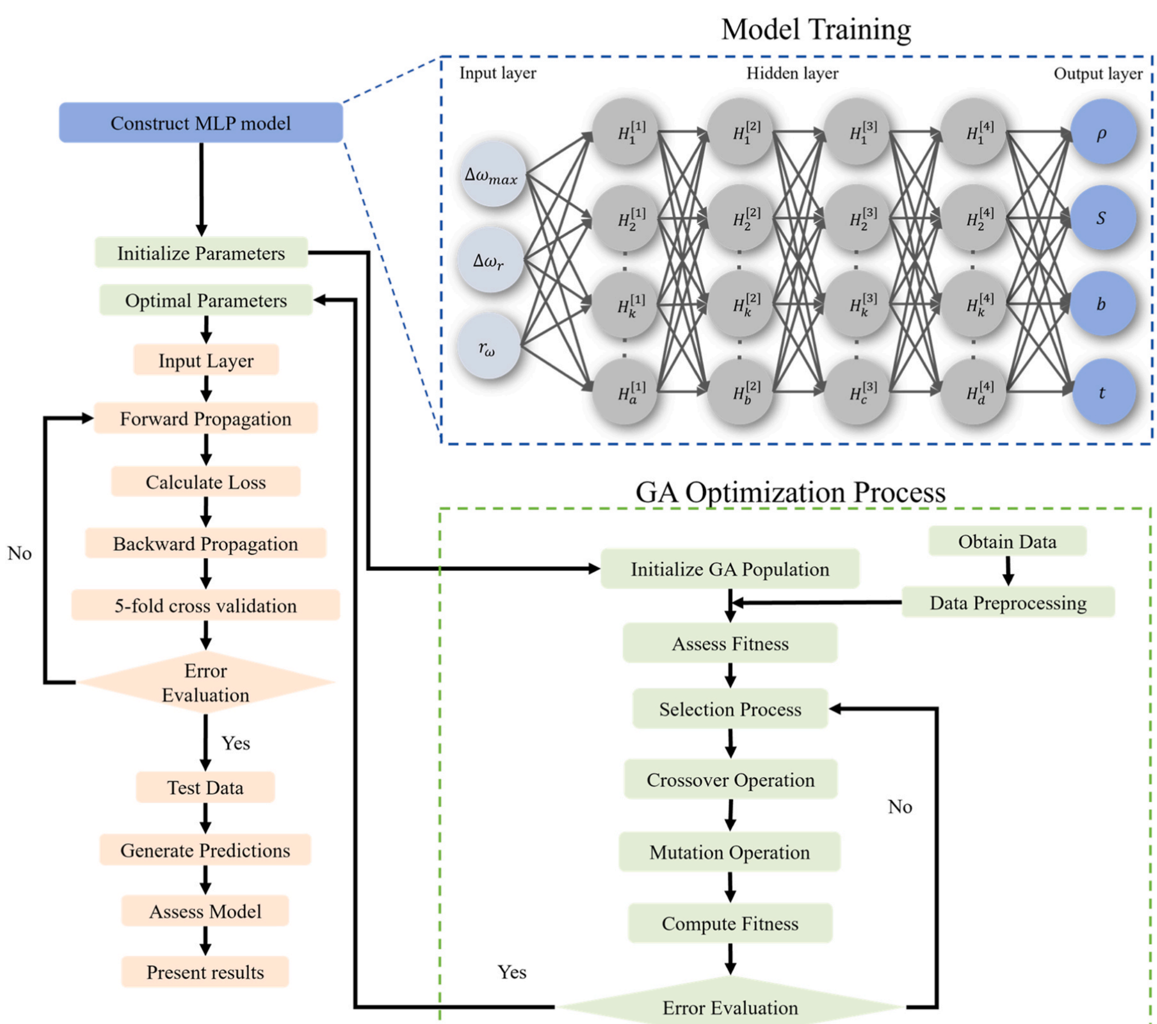

Fig. 8. Architecture of the inverse design model.

evaluate candidate solutions and begins by initializing a population of candidate solutions, each representing a distinct set of parameters for the neural network. The initial population consists of 300 individuals [57], each representing a unique set of parameters. The fitness of each individual (F) is inversely proportional to the MSE, calculated as:

$$F = \frac{1}{MSE}. \tag{14}$$

Throughout 150 generations, the model is progressively refined by the application of selection, crossover (at a rate of 0.7), and mutation (at a rate of 0.15) through the GA. The values were selected based on commonly adopted ranges in previous studies [58,59] and validated through preliminary tuning to ensure stable convergence and sufficient solution diversity for the optimization problem. Once the model's fitness, inversely proportional to the mean squared error (MSE), reaches the predefined threshold, the GA-MLP model undergoes validation. The optimized model is then used to predict the optimal design parameters for metamaterials that meet the specified vibration characteristics, completing the inverse design process from input parameters to desired vibration performance.

4. Results

4.1. Vibration modes

Fig. 9 shows that the ASHM exhibits distinct vibration modes at two different frequencies, 209.21 Hz and 69.615 Hz, each within separate bandgaps. At 209.21 Hz, the outer frame of the metamaterial rotates clockwise, with the acceleration direction aligning with this motion. In contrast, the inner embedded section, particularly the cement part, rotates in the opposite direction, counterclockwise, with its acceleration following the same pattern, as illustrated in Fig. 9(a). This opposing motion between the inner and outer regions demonstrates the occurrence of local resonance, highlighting the metamaterial's unique bandgap properties at this frequency. Similarly, although the acceleration of the inner cement core maintains a consistent counterclockwise direction, there is a variation in the acceleration distribution among the ribs of the outer TPU frame at 69.615 Hz. This difference arises from the relative motion between the ribs, where the coupling effects due to the rib connections cause distinct phase shifts in the vibration response of each rib. Specifically, the relative vibration modes and acceleration directions of the individual ribs within the TPU frame are not identical, leading to local resonance at specific frequencies. This local resonance is a result of the interaction between the inner and outer structures, as well as their non-uniform motion, which enhances the metamaterial's vibration control capabilities and its bandgap properties. This consistent behavior across different frequencies further underscores the metamaterial's ability to achieve localized resonance, confirming the robustness of its bandgap properties under varying conditions. The contrasting rotational directions at these two frequencies highlight the versatility of the metamaterial in manipulating wave propagation and controlling vibration modes.

4.2. Bandgap distribution

Fig. 10 illustrates the bandgap distribution of ASHM within the frequency range of 0–650 Hz. A total of eight bandgaps are identified, with the narrowest bandgap spanning from 209.21 to 216.25 Hz. The lowest-frequency bandgap lies between 71.661 and 201.94 Hz, and it exhibits the widest bandwidth among all bandgaps, measuring 130.279 Hz. This indicates that within this frequency range, the metamaterial effectively inhibits the propagation of elastic waves. The Δw_c, of the maximum bandgap is 0.952, which is significantly larger than that of our previous study [34], demonstrating the metamaterial's ability to maintain effective wave suppression over a relatively lower and border frequency range. In contrast, the narrowest bandgap expands up to 7.04 Hz, with its lower and upper limits at 209.21 Hz and 216.25 Hz, respectively. Although the absorption efficiency is limited within narrow frequency ranges, a pronounced bandgap phenomenon is observed in most other regions, as evidenced by the low-frequency bandgap ratio r_ω reaching 0.515. This indicates that the ASHM effectively impedes mechanical wave transmission across these frequencies, particularly in areas where bandgaps exist.

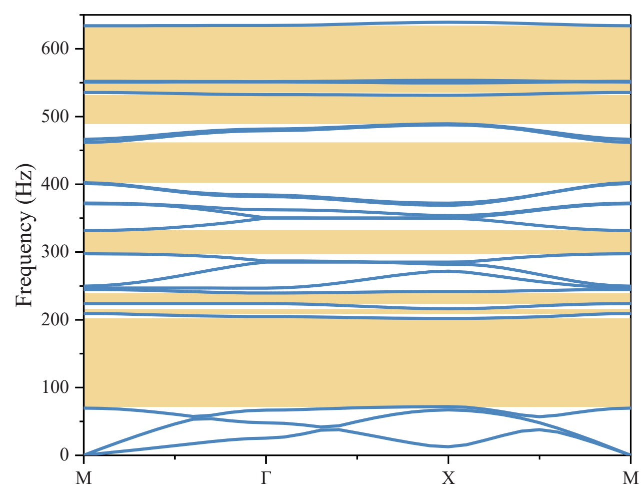

Fig. 10. The bandgap distribution of ASHM.

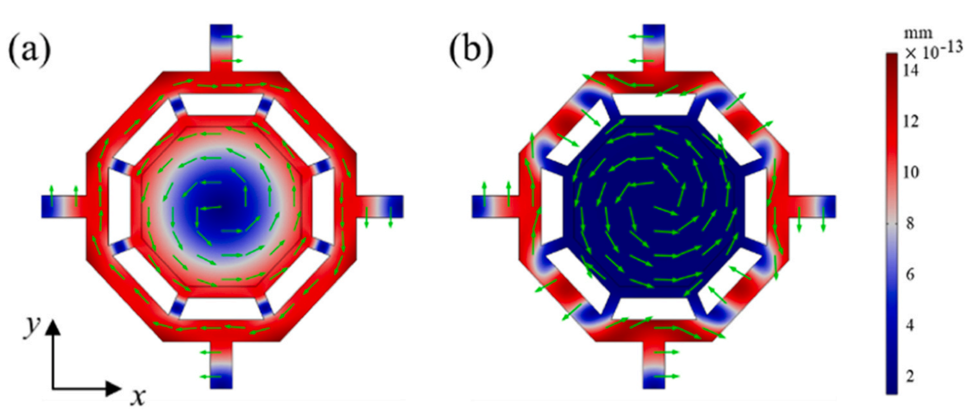

Fig. 9. Vibration mode at different frequencies: (a) 209.21 Hz, and (b) 69.615 Hz.

4.3. Frequency response

Fig. 11 demonstrates both the experimental and numerical results of transmission loss for ASHM. The results indicate a strong correlation between the frequency response of the metamaterial and its bandgap distribution, affirming the reliability of the numerical model in predicting vibration reduction performance. Specifically, within the primary bandgap intervals—such as 71.661–201.94 Hz and 298.15–331.25 Hz—the FRF of the metamaterial reaches minimum values of −12.368 dB and −20.661 dB, respectively, demonstrating maximum transmission loss in these frequency ranges and thereby effectively suppressing the propagation of elastic waves.

However, certain deviations are observed outside the primary bandgap intervals. At specific frequencies, such as 273.13 Hz, the FRF exhibits relatively low values, approximately −13.24 dB. This discrepancy is largely attributed to minor variations in material parameters, as well as structural inconsistencies introduced during fabrication by the FDM technique, which may slightly alter density and elastic modulus. Although these variances have minimal overall impact, they may notably influence local FRF values near bandgap boundaries.

The experimental results exhibit a general alignment with the simulation, particularly within the bandgap frequencies where FRF values remain low, further confirming the notable transmission loss and wave-suppressing capabilities of the metamaterial within these ranges. This attenuation effect not only validates the effectiveness of the metamaterial's design but also confirms its potential for frequency-selective vibration reduction. The observed variations, especially near bandgap edges, are likely influenced by structural imperfections, boundary effects, and the limited number of unit cells used in the experiment. As only a 4 × 4unit configuration was employed, periodic boundary conditions could not be fully satisfied, leading to minor deviations in FRF values within certain frequencies, notably in the 250–300 Hz range. Despite these boundary-induced fluctuations, FRF values generally remain low within the bandgap regions, affirming the metamaterial's effectiveness in attenuating elastic wave propagation across these frequencies.

Fig. 11(b) validates the consistency between the experimental and simulation results. Within the bandgap frequency range (80 Hz and 90 Hz), the propagation of elastic waves in the cementitious metamaterial is inhibited. In contrast, elastic waves at 50, 60, and 70 Hz can penetrate the metamaterial more effectively, exhibiting enhanced propagation capabilities.

The overall consistency between the experimental and simulated results confirms the reliability of the ASHM in achieving transmission loss and vibration reduction. This consistency further substantiates its potential application in tailored vibration control, providing a robust foundation for future design and optimization efforts.

4.4. Forward prediction

Fig. 12 illustrates the prediction results and error distribution of the forward prediction model across three key vibration isolation indicators. The figures show that the machine learning-based predictions align closely with numerical results, with R^2 values exceeding 0.95 for all indicators, demonstrating the model's accuracy and reliability in predicting metamaterial vibration reduction capabilities.

According to Fig. 12 (d) and (g), the model demonstrates low prediction errors for indicator $\Delta w_{\max}$, with a maximum absolute error of 16.68 and RMSE and MAE values of 0.703 and 0.495, indicating robustness and precision in capturing low-frequency attenuation characteristics. Although some samples show relatively larger errors, these remain acceptable given the physical significance of this indicator in representing the maximum bandgap width. For indicator Δw_c, the model achieves an R^2 of 0.964, with predicted values closely matching numerical results. This high level of agreement confirms the model's accuracy in reflecting the bandgap frequency distribution and its impact on performance at specific frequencies. The low RMSE and MAE values for both indicators further verify the model's stable performance across various samples, effectively meeting specific frequency attenuation requirements.

Similarly, the model achieves high prediction accuracy for indicator r_ω, with an R^2 near 0.99 and low RMSE and MSE values of 0.001 and 0.0317. The close alignment between the predicted and actual results for indicator r_ω highlights the model's accuracy in capturing low-frequency bandgap ratios. This precision in predicting indicator r_ω is significant because it directly reflects the model's ability to assess the metamaterial's effectiveness in attenuating low-frequency vibrations.

Overall, these results highlight the forward prediction model's high accuracy and reliability in assessing metamaterial vibration isolation performance. Its strong predictive capability, with R^2 values above 0.95, confirms its effectiveness for accurate evaluation and optimization of vibration isolation systems.

4.5. Inverse design

Fig. 13 presents the inverse design results based on MLP and the GA-MLP models in their test sets. As shown in Fig. 13 (a), the GA-MLP model achieves a high degree of alignment between its predicted values for parameter S and the numerical simulation results, indicating superior predictive accuracy on the test set. Most GA-MLP model errors fall within the range between 0 mm and 2 mm, with an average error of 0.566, as illustrated in Fig. 13 (b), which reflects reliable accuracy across most data points in the test set. Since S represents the length of each unit cell in the ASHM, this average error of 0.566 mm, corresponding to a 5-mm deviation, has minimal impact on the overall design. By

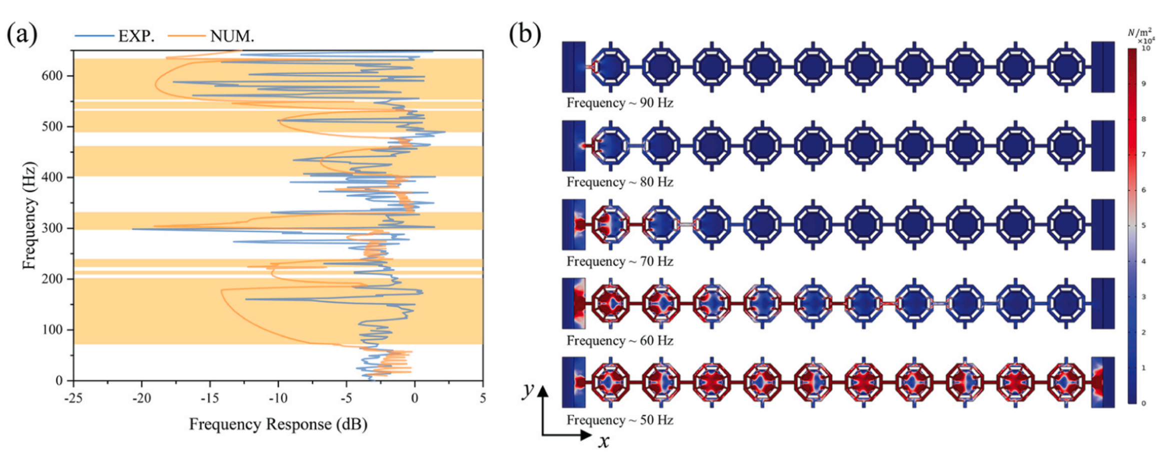

Fig. 11. The experimental and numerical results of frequency response for the ASHM: (a) FRF, (b) vibration modes from 50 Hz to 90 Hz.

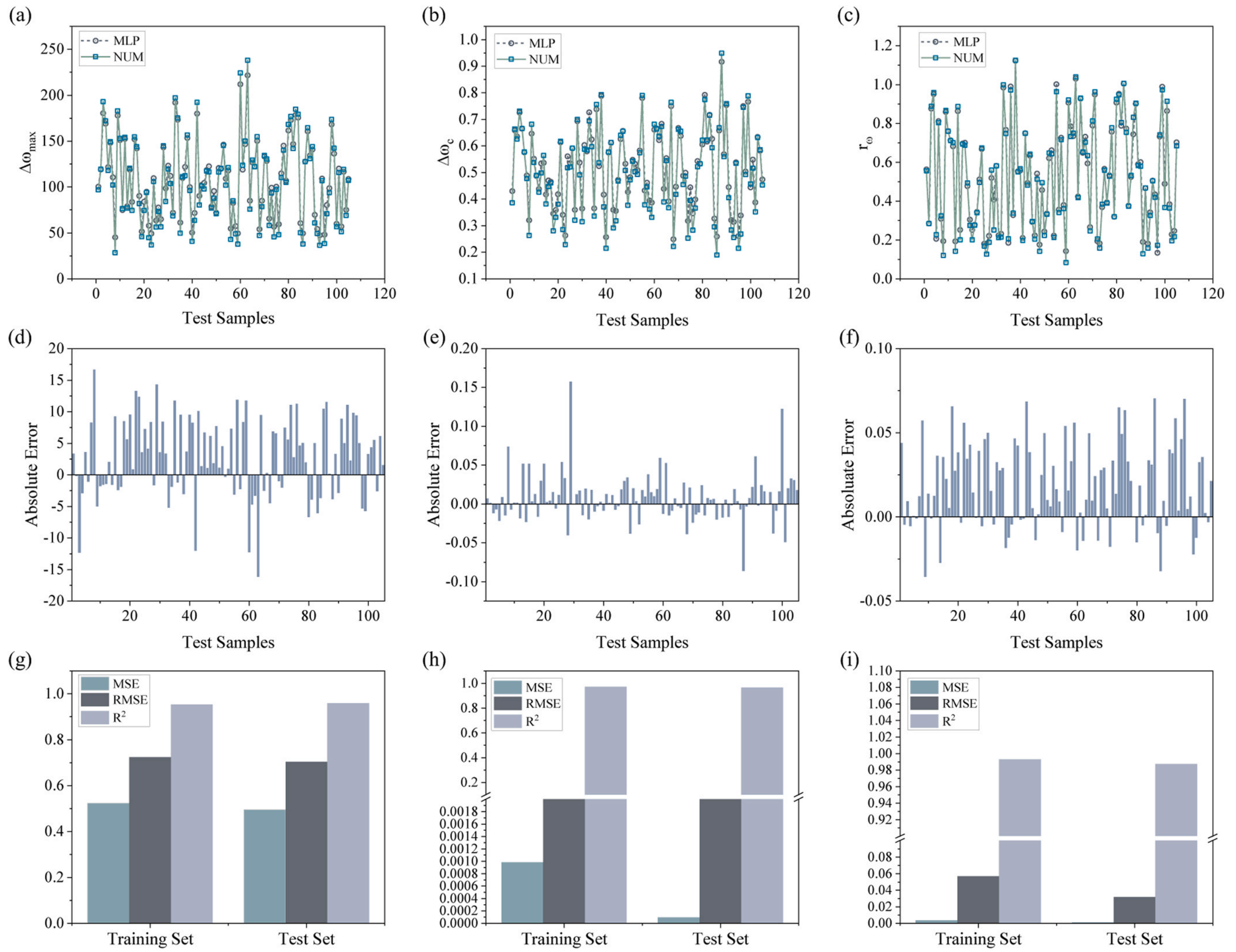

Fig. 12. The prediction results and numerical value of (a) $\Delta w_{\max}$, (b) Δw_c, (c) r_ω, error distributions of (d) $\Delta w_{\max}$, (e) Δw_c, (f) r_ω, and performance metrics of (g) $\Delta w_{\max}$, (h) Δw_c, (i) r_ω.

comparison, predictions from the conventional MLP model exhibit significantly higher absolute error. Specifically, Fig. 13 (c) shows that the MLP model's MSE and RMSE are 1.76 and 1.33, respectively, while the GA-MLP model achieves significantly lower MSE and RMSE values of 0.83 and 0.91. This improvement results primarily from the genetic algorithm, which optimizes the network structure, weights, and biases of the MLP, thereby enhancing generalization capability and predictive accuracy. By globally refining these parameters, the genetic algorithm allows the model to escape local minima and better capture complex patterns within the data. This holistic optimization not only boosts performance on the training set but also ensures greater reliability and precision when the model is applied to new, unseen data.

Similar improvement applies to other geometric parameters, including b and t. As shown in Fig. 13 (f) and (i), the GA-MLP model consistently reaches R^2 values above 0.95 for both parameters, in contrast to the MLP model's lower R^2 values of 0.79, and 0.82, respectively. The GA-MLP model's advantage becomes especially evident for parameter ρ. In Fig. 13 (k), the MLP model's maximum error for ρ exceeds 580, while the GA-MLP model maintains substantially lower RMSE and MSE values of 85.615 and 7330.055, achieving greater accuracy and stability across all parameter predictions.

Overall, the GA-MLP model demonstrates clear advantages over the conventional MLP in the inverse design of metamaterials. Higher accuracy and stable performance across multiple parameters highlight the effectiveness of genetic algorithm optimization. The impact of such optimizations on enhancing network architecture, as refined weight and bias adjustments contribute to greater model reliability, establishing GA-MLP as a more dependable tool for inverse design applications.

5. Discussion

The proposed cementitious metamaterials (ASHM) exhibit bandgap characteristics that allow them to suppress the propagation of elastic waves within specific frequency ranges, effectively functioning as selective wave filters. This capability positions ASHM as a promising solution for vibration mitigation. Previous studies have demonstrated their effectiveness as seismic metamaterials and structural dampers, capable of suppressing low-frequency elastic wave propagation in various applications. In this study, we explore their potential in marine engineering, particularly for offshore floating wind platforms, where vibration control is crucial for long-term structural stability. The data-driven design approach outlined in this study allows precise adjustment of the bandgap distribution in these materials, facilitating adaptive suppression of elastic waves over a broad and varied frequency range. This tunable characteristic is particularly beneficial in complex offshore environments, where ASHM can be engineered to target specific vibrational frequencies caused by waves and wind forces.

Fig. 14 presents a conceptual diagram illustrating the potential

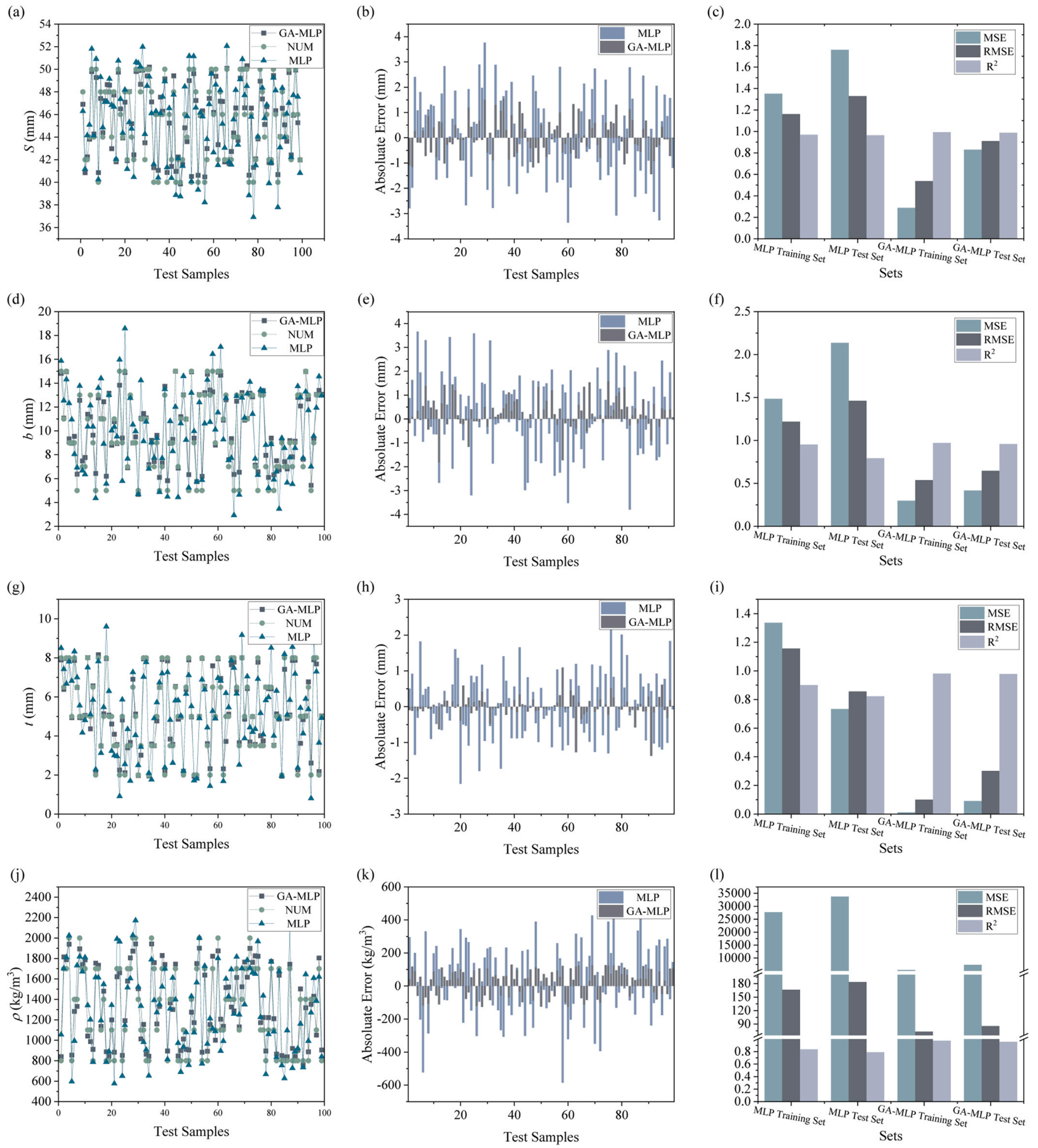

Fig. 13. The prediction results and numerical value of (a) S, (d) b, (g) t, (j) ρ, error distributions of (b) S, (e) b, (h) t, (k) ρ, and performance metrics of (c) S, (f) b, (i) t, (l) ρ.

application of data-driven ASHM in offshore floating wind platforms. The tunable vibration characteristics of cement-based metamaterials allow them to withstand the complex offshore environment, significantly enhancing platform durability and stability. Offshore wind platforms are exposed to challenging environmental factors, including wind, waves, tides, temperature fluctuations, and saltwater corrosion. The combined effects of wind and waves impose continuous vibrations and impact forces on the platform. Specifically, wave-induced forces and wind-induced mechanical vibrations constitute the two primary mechanical loads that affect platform stability and longevity. In floating wind platforms, the turbine base is especially impacted by these forces: periodic wave forces transmit through the platform structure, potentially destabilizing it, while wind-induced vibrations from the turbine blades propagate downward to the base, leading to cumulative fatigue damage to the concrete substructure. Integrating ASHM into the turbine base provides an effective solution for mitigating these external forces.

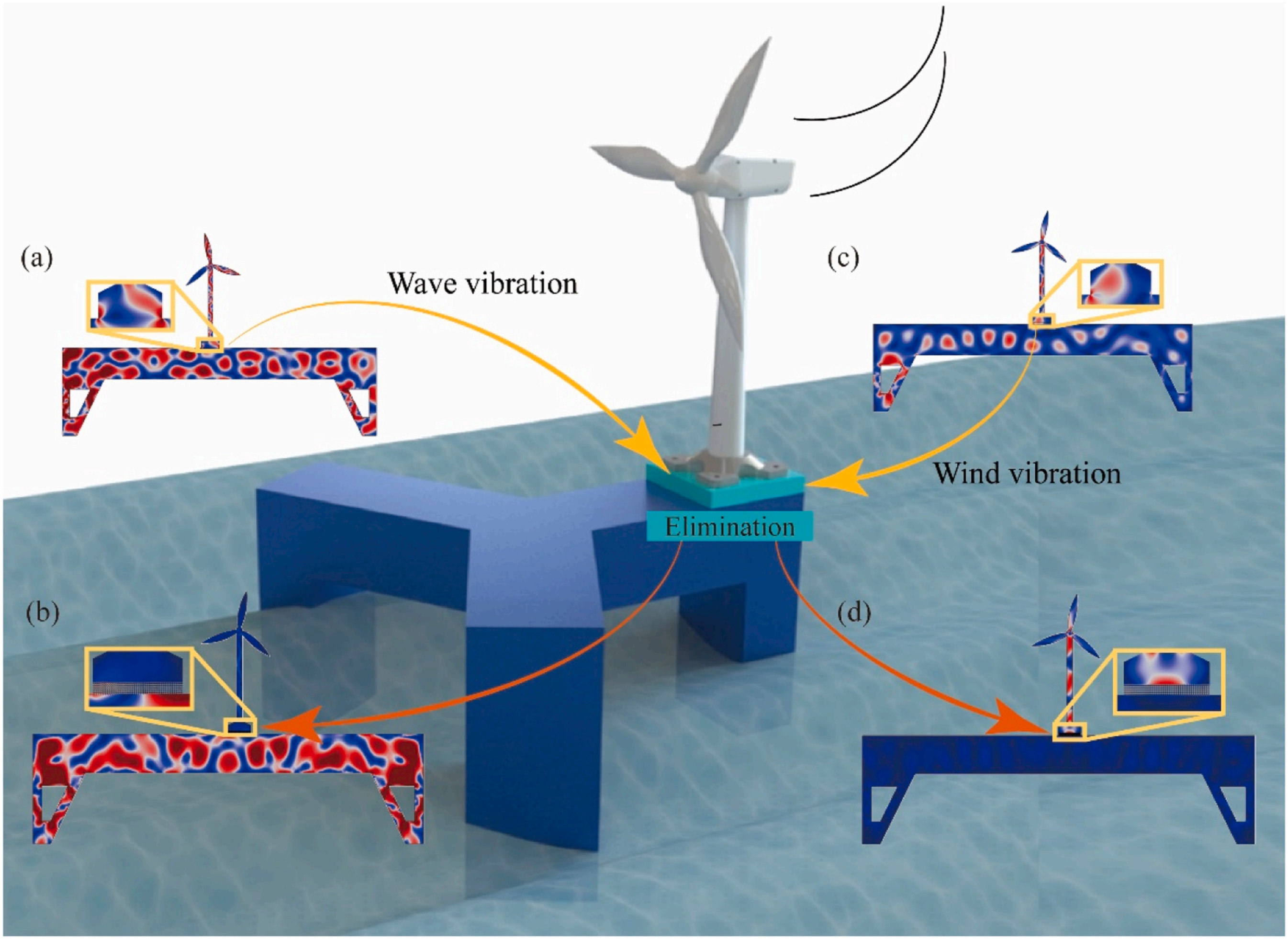

Fig. 14. Conceptual illustration of offshore floating wind platform vibration modes: (a) wave vibration without support; (b) wave vibration with ASHM support; (c) wind vibration without support; (d) wind vibration with ASHM base support.

These ASHM offer two notable advantages:

(a) Blocking wave impact [60]. The tunable structure of the cement-based metamaterial may effectively block and absorb plane wave impacts from waves, dissipating vibrational energy at the base and reducing upward transmission of wave energy (Fig. 14(b)), thereby protecting the stability of the turbine structure. The cementitious metamaterial's design can be adjusted to specific wave frequencies and wavelengths, resulting in exceptional vibration isolation within the frequency range of wave impacts.

(b) Damping wind-induced mechanical vibrations [61]. The cementitious metamaterial base can also serve as an isolator and buffer for mechanical vibrations from the turbine (Fig. 14(d)). When wind-induced vibrations transmit through the turbine structure to the base, the metamaterial can absorb part of this energy, reducing cumulative effects on the concrete substructure and minimizing long-term fatigue damage, thus extending the platform's service life.

On the one hand, the data-driven design of ASHM for turbine bases in offshore wind platforms effectively mitigates wave impacts, suppresses plane wave propagation, and dampens mechanical vibrations. This tunable structure provides crucial support for platform stability, showing strong adaptability and promise for extreme marine conditions. These findings underscore the potential of ASHM in offshore engineering, particularly in addressing vibrational challenges associated with floating wind platforms. To further illustrate their practical applicability, it is valuable to examine specific scenarios where these metamaterials can be effectively implemented. For instance, floating wind platforms, such as those deployed in the Hywind Scotland project, experience persistent hydrodynamic loading, necessitating advanced vibration control strategies. The integration of ASHM into turbine foundations can selectively attenuate wave energy transmission, thereby reducing oscillatory motion, mitigating fatigue-induced damage, and extending the operational lifespan of these structures.

Beyond their application in floating wind platforms, ASHM offers broader advantages in offshore engineering due to their ability to suppress vibrations across a broad frequency spectrum. Their tunability, enabled by data-driven optimization, allows for precise control of structural responses, making them well-suited for environments where dynamic stability is a critical concern. As offshore infrastructure continues to expand into more challenging marine environments, the strategic integration of ASHM could play a pivotal role in enhancing durability, operational efficiency, and sustainability across offshore wind energy, oil and gas platforms, and coastal protection systems.

On the other hand, the proposed application of ASHM for vibration mitigation in offshore floating platforms remains conceptual and has not undergone comprehensive and rigorous validation. Consequently, significant challenges and uncertainties persist in its practical implementation. Offshore floating platforms operate in highly complex environments, characterized by multidirectional wave interactions, tidal variations, temperature fluctuations, and saltwater corrosion. The simplified model presented in this study does not fully capture these environmental complexities, which may significantly influence the performance of ASHM in real-world conditions.

Furthermore, ASHM is currently limited to laboratory-scale research, with its fabrication process optimized for small-scale prototypes. The transition to large-scale manufacturing introduces additional challenges, including material consistency, structural integrity, and production feasibility, all of which require further investigation. In addition, numerical simulations and experimental validations remain constrained in scope. While laboratory tests indicate that ASHM can effectively attenuate elastic wave propagation within specific frequency ranges, the current experimental setup is limited in scale and does not fully replicate the complex dynamic loads and environmental interactions of offshore platforms. To ensure the long-term stability and engineering viability of ASHM, larger-scale experiments, including in-situ testing on operational offshore platforms, are necessary.

6. Conclusion

This study presents an intelligent design strategy for cementitious metamaterials that effectively suppresses low-frequency vibrations by adjusting key geometric parameters and cement density. The designed ASHM demonstrates excellent low-frequency broadband vibration attenuation characteristics. A machine learning framework, employing MLP and GA-MLP networks, enables efficient forward prediction and inverse design of metamaterial properties. This approach rapidly navigates complex design spaces, optimizes structural parameters, and flexibly tunes bandgaps, significantly improving design efficiency and accuracy. It offers a novel solution for low-frequency vibration control and highlights the promising role of machine learning in metamaterial design. The main conclusions of this study can be summarized as follows.

(a) The designed concentric cement-based metamaterial combines cement and polymer to form a phononic crystal with a localized resonance effect. A numerical model of its low-frequency vibrations reveals that different parts of the internal unit cells accelerate in opposing directions. Furthermore, this metamaterial exhibits multiple bandgaps within the low-frequency range (0–650 Hz) despite its compact size, demonstrating relatively high efficiency in elastic wave absorption across this frequency spectrum.

(b) The numerical model for transmission loss in the constructed metamaterial aligns well with experimental FPF results, demonstrating the numerical model's reliability. Minor discrepancies arise due to the limited number of cells in the test sample, which does not fully satisfy Floquet boundary conditions. Nevertheless, the experimental FPF values remain low within the frequency ranges corresponding to simulated bandgaps, confirming effective elastic wave suppression in these bands.

(c) Metamaterials were designed with different combinations of geometric parameters (S, b, t) and density (ρ). Corresponding numerical models were then constructed to determine their vibrational characteristics ($\Delta w_{\max}$, Δw_c, r_ω), resulting in a comprehensive dataset. This dataset enabled the development of a data-driven design method using MLP and GA-MLP models. Results indicate that the machine learning models achieve over 95 % accuracy, demonstrating strong robustness and generalization in predicting both vibrational properties and geometry-material properties.

(d) A GA was introduced to optimize the biases and weights of the original MLP model, forming the GA-MLP predictive model. This model shows significantly improved accuracy in predicting metamaterial geometry and material parameters for specified vibrational performance. Test results indicate MSE and RMSE values of 0.83 and 0.91, respectively, which are markedly lower than those of the MLP model (MSE ~1.76, RMSE ~1.33). In comparison, direct predictions with the MLP model show greater deviation, with an R^2 of approximately 0.79, falling outside the confidence interval.

This study presents a data-driven, cementitious metamaterial with low-frequency vibration suppression and tunable vibrational properties. While initial results are promising, identifying suitable materials and manufacturing processes remains a critical challenge in construction applications. As building materials [62,63] evolve toward low- and even negative-carbon footprints [64], using these innovative materials to create vibration-damping metamaterials offers notable environmental benefits and enhances the multifunctional and vibration-suppressing capabilities of the material. This approach significantly broadens its potential in architectural applications and adds substantial value.

CRediT authorship contribution statement

Liu Yuqing: Validation, Methodology. **Zhang Jianchao:** Validation, Methodology. **Hu Jiayi:** Methodology, Investigation, Formal analysis. **Li Yuanlong:** Methodology, Investigation, Formal analysis. **Gong Zhi:** Methodology, Investigation, Formal analysis. **Li Gui:** Validation, Methodology. **Dong Peng:** Methodology, Investigation, Formal analysis, Conceptualization.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

The authors would like to acknowledge the funding support from the Science, Technology and Innovation Commission of Shenzhen Municipality (grant no. ZDSYS20210623092005017) and Basic and Applied Basic Guangdong-Dongguan Joint Youth Fund Project of Guangdong Province (2023A1515110320).

Data Availability

Data will be made available on request.

References

[1] M. Zhang, T. Guo, R. Zhu, Y. Zong, Z. Liu, W. Xu, Damage identification of seismic-isolated structure based on CAE network using vibration monitoring data, Eng. Struct. 283 (2023) 115873, https://doi.org/10.1016/j.engstruct.2023.115873.

[2] M. Heckl, G. Hauck, R. Wettschureck, Structure-borne sound and vibration from rail traffic, J. Sound Vib. 193 (1996) 175–184, https://doi.org/10.1006/jsvi.1996.0257.

[3] H. Iwabuki, S. Fukada, T. Osafune, M. Shimura, E. Sasaki, Contribution of large-vehicle vibration and bridge vibration to low-frequency noise generation characteristics, Appl. Acoust. 155 (2019) 150–166, https://doi.org/10.1016/j.apacoust.2019.05.011.

[4] P.K. Singh, M.P. Roy, Damage to surface structures due to blast vibration, Int. J. Rock. Mech. Min. Sci. 47 (2010) 949–961, https://doi.org/10.1016/j.ijrmms.2010.06.010.

[5] H. Hao, K. Bi, W. Chen, T.M. Pham, J. Li, Towards next generation design of sustainable, durable, multi-hazard resistant, resilient, and smart civil engineering structures, Eng. Struct. 277 (2023) 115477, https://doi.org/10.1016/j.engstruct.2022.115477.

[6] K. Kamei, M.A. Khan, Current challenges in modelling vibrational fatigue and fracture of structures: a review, J. Braz. Soc. Mech. Sci. Eng. 43 (2021) 77, https://doi.org/10.1007/s40430-020-02777-6.

[7] C. Baliatsas, I. van Kamp, R. van Poll, J. Yzermans, Health effects from low-frequency noise and infrasound in the general population: Is it time to listen? A systematic review of observational studies, Sci. Total Environ. 557–558 (2016) 163–169, https://doi.org/10.1016/j.scitotenv.2016.03.065.

[8] L. Wang, W. Shi, Q. Zhang, Y. Zhou, Study on adaptive-passive multiple tuned mass damper with variable mass for a large-span floor structure, Eng. Struct. 209 (2020) 110010, https://doi.org/10.1016/j.engstruct.2019.110010.
[9] X. Sagartzazu, L. Hervella-Nieto, J.M. Pagalday, Review in sound absorbing materials, Arch. Comput. Methods Eng. 15 (2008) 311–342, https://doi.org/10.1007/s11831-008-9022-1.
[10] K.P. Biligiri, Effect of pavement materials' damping properties on tyre/road noise characteristics, Constr. Build. Mater. 49 (2013) 223–232, https://doi.org/10.1016/j.conbuildmat.2013.08.016.
[11] L. Cao, Q. Fu, Y. Si, B. Ding, J. Yu, Porous materials for sound absorption, Compos. Commun. 10 (2018) 25–35, https://doi.org/10.1016/j.coco.2018.05.001.
[12] X. Qin, W. Yang, Z. Zhang, Z. Chen, Research on the design and noise reduction performance of periodic noise barriers based on nested structure, J. Clean. Prod. 476 (2024) 143708, https://doi.org/10.1016/j.jclepro.2024.143708.
[13] Y. Fu, I.I. Kabir, G.H. Yeoh, Z. Peng, A review on polymer-based materials for underwater sound absorption, Polym. Test. 96 (2021) 107115, https://doi.org/10.1016/j.polymertesting.2021.107115.
[14] G. Fu, Y. Quan, M. Gu, Z. Huang, C. Feng, Dynamic performance evaluation of a 492 m super high-rise building with active tuned mass dampers during four consecutive landfall typhoons within a month, J. Build. Eng. 61 (2022) 105259, https://doi.org/10.1016/j.jobe.2022.105259.
[15] H.Q. Cao, Combined Tuned Mass Dampers for structural vibration control, Int. J. Non-Linear Mech. 157 (2023) 104550, https://doi.org/10.1016/j.ijnonlinmec.2023.104550.
[16] Y. Zhang, T. Schauer, A. Bleicher, Optimized passive/semi-active vibration control using distributed-multiple tuned facade damping system in tall buildings, J. Build. Eng. 52 (2022) 104416, https://doi.org/10.1016/j.jobe.2022.104416.
[17] L. Koutsoloukas, N. Nikitas, P. Aristidou, Passive, semi-active, active and hybrid mass dampers: A literature review with associated applications on building-like structures, Dev. Built Environ. 12 (2022) 100094, https://doi.org/10.1016/j.dibe.2022.100094.
[18] S.A. Cummer, J. Christensen, A. Alù, Controlling sound with acoustic metamaterials, Nat. Rev. Mater. 1 (2016) 16001, https://doi.org/10.1038/natrevmats.2016.1.
[19] S. Brûlé, S. Enoch, S. Guenneau, Emergence of seismic metamaterials: Current state and future perspectives, Phys. Lett. A 384 (2020) 126034, https://doi.org/10.1016/j.physleta.2019.126034.
[20] Z. Guo, J. Wen, Y. Shen, G. Hu, G. Jiang, Band folding induced broadband vibration suppression of star-shaped metamaterials: Theory and experiment, Thin-Walled Struct. 198 (2024) 111756, https://doi.org/10.1016/j.tws.2024.111756.
[21] Q. Lu, C. Liu, P. Wang, Band gap enhancement and vibration reduction of functionally graded sandwich metastructure beam, Compos. Struct. 292 (2022) 115650, https://doi.org/10.1016/j.compstruct.2022.115650.
[22] W. Chen, Y.X. Hao, W. Zhang, S.W. Yang, Y.T. Cao, Vibration Isolation Performance of a Novel Metamaterials Sandwich Cylindrical Panel by Locally Resonant Band Gap, J. Vib. Eng. Technol. 12 (2024) 6121–6136, https://doi.org/10.1007/s42417-023-01242-7.
[23] A. Arjunan, A. Baroutaji, J. Robinson, A. Vance, A. Arafat, Acoustic metamaterials for sound absorption and insulation in buildings, Build. Environ. 251 (2024) 111250, https://doi.org/10.1016/j.buildenv.2024.111250.
[24] Y.M. Luo, T.T. Huang, Y. Zhang, H.H. Xu, Y.M. Xie, X. Ren, Novel meter-scale seismic metamaterial with low-frequency wide bandgap for Lamb waves, Eng. Struct. 275 (2023) 115321, https://doi.org/10.1016/j.engstruct.2022.115321.
[25] Z. Tao, X. Ren, A.G. Zhao, L. Sun, Y. Zhang, W. Jiang, D. Han, X.Y. Zhang, Y.M. Xie, A novel auxetic acoustic metamaterial plate with tunable bandgap, Int. J. Mech. Sci. 226 (2022) 107414, https://doi.org/10.1016/j.ijmecsci.2022.107414.
[26] A. Colombi, R. Zaccherini, G. Aguzzi, A. Palermo, E. Chatzi, Mitigation of seismic waves: Metabarriers and metafoundations bench tested, J. Sound Vib. 485 (2020) 115537, https://doi.org/10.1016/j.jsv.2020.115537.
[27] X. An, G. Sun, X. Yuan, Y. Tian, X. Hou, H. Fan, Design of Kagome lattice composite sandwich metastructures with high load bearing and low frequency vibration reduction properties, Compos. A: Appl. Sci. Manuf. 174 (2023) 107716, https://doi.org/10.1016/j.compositesa.2023.107716.
[28] Z.-K. Guo, G. Hu, V. Sorokin, Y. Yang, L. Tang, Sound transmission through sandwich plate with hourglass lattice truss core, Jnl Sandw. Struct. Mater. 23 (2021) 1902–1928, https://doi.org/10.1177/1099636220906819.
[29] E. Tezsezen, D. Yigci, A. Ahmadpour, S. Tasoglu, AI-Based Metamaterial Design, ACS Appl. Mater. Interfaces (2024), https://doi.org/10.1021/acsami.4c04486.
[30] P. Dong, R. Hou, J. Hu, C. Lin, Y. Liu, L. Qin, Novel self-similar re-entrant auxetic metamaterials (SREAM): Design, mechanical property, and geometric optimization, Polym. Test. 122 (2023) 108015, https://doi.org/10.1016/j.polymertesting.2023.108015.
[31] J. Hu, P. Dong, R. Hou, J. Cao, S. Sadeghzade, H. Yuan, Functionally graded IWP reinforced cementitious composites: Design, fabrication, and the enhanced ductility, Thin-Walled Struct. 192 (2023) 111199, https://doi.org/10.1016/j.tws.2023.111199.
[32] Z. Zhang, Q. Wen, P. Li, H. Hu, Application of double arrowhead auxetic honeycomb structure in displacement measurement, Sens. Actuators A: Phys. 333 (2022) 113218, https://doi.org/10.1016/j.sna.2021.113218.
[33] H. Masood, T. Sirojan, C.Y. Toe, P.V. Kumar, Y. Haghshenas, P.H.-L. Sit, R. Amal, V. Sethu, W.Y. Teoh, Enhancing prediction accuracy of physical band gaps in semiconductor materials, Cell Rep. Phys. Sci. 4 (2023) 101555, https://doi.org/10.1016/j.xcrp.2023.101555.
[34] P. Dong, J. Hu, Z. Gong, Y. Li, J. Liu, J. Cao, C. Lin, W. Ding, Design strategies of cementitious metamaterials (CMs) with tunable bandgaps: Density customization and geometric optimization, J. Build. Eng. 95 (2024) 110132, https://doi.org/10.1016/j.jobe.2024.110132.
[35] J. Hu, P. Dong, J. Cao, Z. Gong, R. Hou, H. Yuan, Dynamic bandgap responsiveness of modified re-entrant metamaterials in thermal-mechanical field, J. Mater. Res. Technol. 30 (2024) 7328–7339, https://doi.org/10.1016/j.jmrt.2024.05.135.
[36] H.-W. Dong, C. Shen, Z. Liu, S.-D. Zhao, Z. Ren, C.-X. Liu, X. He, S.A. Cummer, Y.-S. Wang, D. Fang, L. Cheng, Inverse design of phononic meta-structured materials, Mater. Today (2024), https://doi.org/10.1016/j.mattod.2024.09.012.
[37] T. Park, J. Song, J. Jeong, S. Kang, J. Kim, J. Won, J. Han, K. Min, Interpretable machine learning boosting the discovery of targeted organometallic compounds with optimal bandgap, Mater. Today Adv. 23 (2024) 100520, https://doi.org/10.1016/j.mtadv.2024.100520.
[38] C.-X. Liu, G.-L. Yu, Inverse design of locally resonant metabarrier by deep learning with a rule-based topology dataset, Comput. Methods Appl. Mech. Eng. 394 (2022) 114925, https://doi.org/10.1016/j.cma.2022.114925.
[39] H. Shi, K. Zhang, X. Liu, L. Qi, F. Hong, Z. Deng, Vibration suppression of a meta-structure with hybridization of Kresling origami and Yoshimura origami, Mech. Syst. Signal Process 224 (2025) 111987, https://doi.org/10.1016/j.ymssp.2024.111987.
[40] W. Jiang, Y. Zhu, G. Yin, H. Lu, L. Xie, M. Yin, Dispersion relation prediction and structure inverse design of elastic metamaterials via deep learning, Mater. Today Phys. 22 (2022) 100616, https://doi.org/10.1016/j.mtphys.2022.100616.
[41] A. Merchant, S. Batzner, S.S. Schoenholz, M. Aykol, G. Cheon, E.D. Cubuk, Scaling deep learning for materials discovery, Nature 624 (2023) 80–85, https://doi.org/10.1038/s41586-023-06735-9.
[42] Y. Jin, S. Zeng, Z. Wen, L. He, Y. Li, Y. Li, Deep-subwavelength lightweight metastructures for low-frequency vibration isolation, Mater. Des. 215 (2022) 110499, https://doi.org/10.1016/j.matdes.2022.110499.
[43] D. Chen, Y. Li, Z. Pan, X. Li, T. Xu, X. Li, Low-frequency vibration bandgaps and deep learning-based intelligent design method of Y-shaped core sandwich metabeams, Compos. Struct. 341 (2024) 118214, https://doi.org/10.1016/j.compstruct.2024.118214.
[44] K. Zhang, Y. Guo, X. Liu, F. Hong, X. Hou, Z. Deng, Deep learning-based inverse design of lattice metamaterials for tuning bandgap, Extrem. Mech. Lett. 69 (2024) 102165, https://doi.org/10.1016/j.eml.2024.102165.
[45] R. Hou, P. Dong, J. Hu, Z. Gong, S. Sadeghzade, J. Cao, H. Yuan, An optimized lozenge-chiral auxetic metamaterial with tunable auxeticity and stiffness, Mater. Des. 237 (2024) 112530, https://doi.org/10.1016/j.matdes.2023.112530.
[46] J. Morris, W. Wang, T. Plaisted, C.J. Hansen, A.V. Amirkhizi, Optimizing graded metamaterials via genetic algorithm to control energy transmission, Int. J. Mech. Sci. 263 (2024) 108775, https://doi.org/10.1016/j.ijmecsci.2023.108775.
[47] Z. Li, X. Hou, Y. Ke, M. Tao, Topology optimization with a genetic algorithm for the structural design of composite porous acoustic materials, Appl. Acoust. 197 (2022) 108917, https://doi.org/10.1016/j.apacoust.2022.108917.
[48] T. Jiang, S. Han, Q. Han, C. Li, Design and optimization of the dual-functional lattice-origami metamaterials, Compos. Struct. 327 (2024) 117670, https://doi.org/10.1016/j.compstruct.2023.117670.
[49] N. Di Cesare, D. Chamoret, M. Domaszewski, Optimum topological design of negative permeability dielectric metamaterial using a new binary particle swarm algorithm, Adv. Eng. Softw. 101 (2016) 149–159, https://doi.org/10.1016/j.advengsoft.2015.10.011.
[50] J. Liu, W.-Z. Ma, W. Chen, Y.-S. Chen, X.-C. Deng, Y. Gu, A Metamaterial Absorber Based on Particle Swarm Optimization Suitable for Earth's Atmospheric Transparency Window, IEEE Access 9 (2021) 92941–92951, https://doi.org/10.1109/ACCESS.2021.3089895.
[51] J. Chen, J. Huang, M. An, P. Hu, Y. Xie, J. Wu, Y. Chen, Application of machine learning on the design of acoustic metamaterials and phonon crystals: a review, Smart Mater. Struct. 33 (2024) 073001, https://doi.org/10.1088/1361-665X/ad51bc.
[52] H. Zhang, Y. Wang, H. Zhao, K. Lu, D. Yu, J. Wen, Accelerated topological design of metaporous materials of broadband sound absorption performance by generative adversarial networks, Mater. Des. 207 (2021) 109855, https://doi.org/10.1016/j.matdes.2021.109855.
[53] S. Han, Q. Han, C. Li, Deep-learning-based inverse design of phononic crystals for anticipated wave attenuation, J. Appl. Phys. 132 (2022) 154901, https://doi.org/10.1063/5.0111182.
[54] L. Liu, L.-X. Xie, W. Huang, X.J. Zhang, M.-H. Lu, Y.-F. Chen, Broadband acoustic absorbing metamaterial via deep learning approach, Appl. Phys. Lett. 120 (2022) 251701, https://doi.org/10.1063/5.0097696.
[55] X. Li, S. Cheng, H. Yang, Q. Yan, B. Wang, Y. Sun, Q. Ding, H. Yan, Z. Fan, Y. Xin, Low-Frequency Vibration and Noise Reduction and Wave Propagation Properties of Hexagonal Hybrid Metamaterials with Local Resonance Strips, Phys. Status Solidi (b) 260 (2023) 2200503, https://doi.org/10.1002/pssb.202200503.
[56] H. Gao, Q. Yan, X. Liu, Y. Zhang, Y. Sun, Q. Ding, L. Wang, J. Xu, H. Yan, Low-Frequency Bandgaps of the Lightweight Single-Phase Acoustic Metamaterials with Locally Resonant Archimedean Spirals, Materials 15 (2022) 373, https://doi.org/10.3390/ma15010373.
[57] P. Dong, Z. Gong, Y. Li, J. Hu, J. Cao, R. Hou, Tensile behavior analysis and intelligent geometric optimization of concentric re-entrant auxetic metamaterials (CREAMs), Mater. Today Commun. 44 (2025) 112203, https://doi.org/10.1016/j.mtcomm.2025.112203.
[58] J. Hu, Z. Gong, Y. Li, P. Dong, H. Yuan, Design and optimization of a mechanical metamaterial featuring dual tunability in auxeticity and bandgap modulation, Compos. Struct. 360 (2025) 119050, https://doi.org/10.1016/j.compstruct.2025.119050.

[59] S. Ding, C. Su, J. Yu, An optimizing BP neural network algorithm based on genetic algorithm, Artif. Intell. Rev. 36 (2011) 153–162, https://doi.org/10.1007/s10462-011-9208-z.
[60] F. Zeng, N. Zhang, G. Huang, Q. Gu, M. He, Dynamic response of floating offshore wind turbines under freak waves with large crest and deep trough, Energy 278 (2023) 127679, https://doi.org/10.1016/j.energy.2023.127679.
[61] N. Khodaie, Parametric study of wind-induced vibration control of tall buildings using TMD and TLCD systems, Structures 57 (2023) 105126, https://doi.org/10.1016/j.istruc.2023.105126.
[62] P. Dong, J. Liu, H. Wang, H. Yuan, Q. Wang, Sustainable municipal solid waste incineration fly ash (MSWIFA) alkali-activated materials in construction: Fabrication and performance, Nanotechnologies Constr. Sci. Internet-J. 14 (2022) 43–52, https://doi.org/10.15828/2075-8545-2022-14-1-43-52.
[63] J. Wei, J. Liu, K.H. Khayat, W.-J. Long, Synergistic Effect of HEDP.4Na and Different Induced Pouring Angles on Mechanical Properties of Fiber-Reinforced Alkali-Activated Slag Composites, Fibers 11 (2023) 23, https://doi.org/10.3390/fib11030023.
[64] M.G. Fikru, T. Shen, J. Brodmann, H. Ma, A strategic insight into the market for carbon management capacity, Sustain. Futures 8 (2024) 100374, https://doi.org/10.1016/j.sftr.2024.100374.